-bit ... es for ... nonze...

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – NUMERICAL HACKATHON

Course Code:	ITA0309	**Course Title:**	Mobile computing
CO Assessed:	CO3 – Precision (BL3)	**Assessment:**	Assessment Tool 3 – HACKATHON
Weightage:	30 % of CIA	**Total Marks:**	100
CO3 Statement:	*Implement wireless networking and Mobile IP concepts for routing in cellular and ad hoc networks. (BTL 3)*		
Student Name:	DINDU PUJITHA	**Reg. No.:**	192373037
Section / Batch:	slot D - 2023	**Date:**	11/08/2026

Code of Conduct

I DINDU PUJITHA (192373037) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: D pujitha.

Instructions

- Duration: 60 minutes
- Number of questions : 10

Faculty In-charge	Course Coordinator	Head of Department
Dr. Terrance fedrick fernandez.	Dr. Terrance fedrick fernandez.	
Signature:	Signature:	Signature:
Date: 11/08/26	Date: 11/08/26	Date:

90/100

a) Given :-

Data packet size = 500 bytes

Tunel overhead = 20 bytes

Additional overhead = 20 bytes.

b) so, the total packet size after tuneling is.

500 + 20 = 520 bytes.

2. Maximum Theoretical data rate :

QAM = Quadratue Amplitude Modulaton.

Use :- QAM allows more bits to be transmitted Each symbol by changing two properties of signal.

Given :-

Bandwidth = 20 MHz

Modulation = 64 QAM

Each symbol carriers = 6 bits.

Formula :-

Data rate = Bandwidth × Bits per symbol

$= 20 \times 10^6 \times 6$

$= 120 \times 10^6$ bits/sec.

Maximum theoretical data rate = 120 Mbps.

3. Routing Table Memory Requirement :-

Given:

No. of routing entries = 50

Memory reauired per Entry = 32 bytes

Formula:-

Total memory = No. of Entries x Memory per Entry.

= 50 x 32

= 1600 bytes.

convert to KB:

1600 ÷ 1024 = 1.5625 KB

Total memory required = 1600 bytes ≈ 1.50 KB

4. Distance Travelled between routing updates:-

Given:-

vehicle speed = 10 m/s

Routing update interval = 2 seconds.

formula:

Distance = Speed x Time

= 10 x 2

= 20 meters.

Distance travelled b/w consecutive updates = 20 M.

5. DSR Route Discovery:-

Given :-

No. of intermediate nodes = 5

Each nodes forwards the route request only once.

The route request packet is forwarded through each of the 5 intermediate nodes.

So, the no. of route request packets generated is = 5

Total Route request packets = 5 + 1 = 6

1- orginating

5. intermediate.

Mobile IP subnet Addressing:

subnet mask:-

subnet mask is a number used to divide an IP address into network part and host part.

Examples:-

IPv4: 192.168.1.10

→ uses 32-bit addresses

IPv5 : There is no standard IPv5 used on the Internet

The version after IPv4 is IPv6

→ not officially used

The question is partly hidden by the PDF controls, but the visible information gives:

subnet mask : 255.255.255.240

This subnet mask corresponds to /28

Step 1: Find the no. of host bits

= 32-28 = 4 host bits.

Step 2: calculate Total addresses

= 2^4 = 16 addresses.

Step 3: calculate usable host addresses. two addresses are reserved:

1 for Network ID

1 for Broadcast address.

1- orginating

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1 for Network ID

1 for Broadcast address.

$= 16 - 2 = 14$

subnet mask = 255.255.255.240 (/28)

Total addresses per subnet = 16

usable host addresses per subnet = 14

7. Spectral Efficiency:-

Given:-

Bandwidth = 20 MHz

symbol rate = 1 MSps (1 Million symbols/second)

formula:

$$\text{spectral Efficiency} = \frac{\text{symbol rate}}{\text{Bandwidth}}$$

$$= \frac{1 \times 10^{6}}{20 \times 10^{6}}$$

$= 0.05$ symbols/Hz

spectral Efficiency = 0.05 symbols/Hz.

8. Network Density:-

Given:-

No. of sensor nodes = 50

Deployment area = 1 km^2

$$\text{Network density} = \frac{\text{No. of nodes}}{\text{Area}}$$

$= 50/1$

$= 50$ nodes/km^2

Network density = 50 nodes/km^2

Doppler shift :-

Given:

Train speed, $v = 30 m/s$

carrier frequency, $f_c = 900$ MHz $= 900 \times 10^6$ Hz

speed of light, $c = 3 \times 10^8 m/s$

$$f_d = v f_c / c$$

$$f_d = \frac{30 \times 900 \times 10^6}{3 \times 10^8}$$

$$= 90 \times 10^2$$

$$f_d = 9000 Hz$$

Doppler shift = 9000 Hz = 9 kHz.

10. packet Transmission Time :-

Given :-

Data rate = 54 Mbps.

packet size = 1500 bytes.

step1 : convert packet size into bits

$= 1500 \times 8$

$= 12000$ bits

Step 2: Use the formula

$$\text{Transmission Time} = \frac{\text{packet size(bits)}}{\text{Data rate (bits/sec)}}$$

$$= \frac{12000}{54 \times 10^6}$$

$= 0.0002222$ seconds.

convert to milliseconds:

$$0.0002222 \times 1000 = 0.2222\text{ ms}$$

Transmission time ≈ 0.222 ms